Ramsey's theorem for pairs preserves a finite level induction or collection scheme

Quentin Le Houérou, Ludovic Levy Patey, Keita Yokoyama

arxiv:2402.11616v3; CC BY 4.0

Editorial note. The counted unit is the article's own Proposition prop:rt22-sig2 at source lines 1506-1514 together with the proof environment that begins on the next line (1515-1531); the pairing is the ordinary adjacent one.

Editorial note. The three context spans are the statements of the COH, SEM and SADS propositions that the proof applies; they are counted by sibling cards of this lane and are retained here as context only, their proofs are not retained.

Editorial note. The counted proof carries the author citation of Fiori-Carones et al. The entry is supplied as a task-directory fragment taken verbatim from the bibtex-formatted entry for that key (the bibtex backend itself cannot be used here because the harness runs both xelatex passes before bibtex, so a generated bbl is never read back).

Editorial note. The retained preamble reproduces verbatim the author's document class at source line 2, his theorem declarations at source lines 16-25, his macro definitions at source lines 46-88 and 91, and his natural-operation macros at source lines 104-107, and it retains mathabx and cleveref; the stix-dependent LS2 block and the inputenc load with the utf8x option are not retained.

Editorial note. The remarks at source lines 1495-1499 and 1533-1538 carry citations or refer to unnumbered items; they are not counted here and are not retained.

Allowed standard prerequisites (editorial).

- The three forcing propositions chained by the proof. The COH proposition produces a cohesive set controlling the jump; the SEM proposition produces a transitive subtournament with the same control; the SADS proposition produces a homogeneous set; their statements are retained with this card at source lines 461-468, 760-767 and 1102-1109.. The proof applies them in sequence, each time with alpha equal to one, zero or one so that the well-foundedness requirement is met, and reads off the jump control and the homogeneity.
- The omega-extension result of Fiori-Carones et al. From a set P one obtains a P-computable countable coded model of WKL_0-star containing

the jump of Y, hence an intermediate set P_0 below P and above the jump of Y whose model satisfies the same finite-level scheme; the citation is carried in the proof and its entry is attached with this card.. It supplies the intermediate set P_0 used by the first application, and the fact that both models satisfy the scheme at level n minus one.

- Complexity transfer between the model and its extension. Every Sigma^0_n formula over the extension by G_0 is Sigma^0_{n-1} over the model of P_0, so satisfaction of the scheme at level n minus one in the latter gives satisfaction of the scheme at level n in the former; the proof uses this twice.. It is how the proof raises the scheme from level n minus one to level n at the first and second steps.

Proposition 0.1. *Consider a countable model $\mathcal{M} = (M, S) \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\alpha)$ topped by a set $Y \in S$, where $\alpha \leqslant \epsilon_0$ is an ordinal. Then for every countable sequence of sets $\vec{R} \in S$ and every $P \gg Y'$ such that $\mathcal{M}[P] \models \mathsf{RCA}^*_0$, there exists $G \subseteq M$ such that*

1. *G is $\vec{R}$-cohesive ;*
2. *$P \geqslant_T (G \oplus Y)'$;*
3. *$\mathcal{M}[G] \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\alpha)$.*

Proposition 0.2. *Consider a countable model $\mathcal{M} = (M, S) \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\sup_{\beta<\alpha} \omega^{\beta\times\omega})$ topped by a set $Y \in S$, where $\alpha \in M$ is a non-zero ordinal $\leqslant \epsilon_0$. Then for every infinite stable tournament R in S and every set $P \gg Y'$ such that $\mathcal{M}[P] \models \mathsf{RCA}^*_0$, there exists $G \subseteq M$ such that*

1. *G is an M-infinite transitive subtournament ;*
2. *$P \geqslant_T (G \oplus Y)'$;*
3. *$\mathcal{M}[G] \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\alpha)$.*

Proposition 0.3. *Consider a countable model $\mathcal{M} = (M, S) \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\sup_{\beta<\alpha} \omega^{\beta\times\omega})$ topped by a set $Y \in S$, where $\alpha \in M$ is an ordinal $\leqslant \epsilon_0$. Then for every linear order $L = (M, \leqslant_L) \in S$ of order type $\omega + \omega^*$, there exists $G \subseteq M$ such that*

1. *G is an M-infinite ascending or descending sequence for $\leqslant_L$;*
2. *$Y' \geqslant_T (G \oplus Y)'$;*
3. *$\mathcal{M}[G] \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2 + \mathsf{WF}(\alpha)$.*

Proposition 0.4. *Fix $n \geqslant 2$ and let T_n be either $\mathsf{I}\Sigma^0_n$, or $\mathsf{B}\Sigma^0_{n+1}$. Consider a countable model $\mathcal{M} = (M, S) \models \mathsf{RCA}_0 + T_{n-1}$ topped by a set $Y \in S$. Then, for every coloring $f : [M]^2 \to 2$ in S and every set $P \gg Y'$ such that $\mathcal{M}[P] \models \mathsf{RCA}_0$, there exists $G \subseteq M$ such that*

1. *G is an M-infinite f-homogeneous set ;*

2. *$P \geqslant_T (G \oplus Y)'$;*

3. *$\mathcal{M}[G] \models \mathsf{RCA}_0 + T_n$.*

Proof. By Fiori-Carones et al [1, Lemma 3.2], there exists a P-computable countable coded model of WKL_0^* containing Y'. In particular, there exists a set $P_0 \subseteq M$ such that $P \gg P_0 \gg Y'$. Moreover, $\mathcal{M}[P]$ and $\mathcal{M}[P_0]$ both satisfy T_{n-1}.

Let $\vec{R} = R_0, R_1, \ldots$ be defined by $R_x = \{y : f(x, y) = 1\}$. By Proposition 0.1 applied to $\mathcal{M}$, P_0 and $\alpha = 1$, there exists some M-infinite $\vec{R}$-cohesive set $G_0 \subseteq M$ such that $P_0 \geqslant_T (G_0 \oplus Y)'$ and $\mathcal{M}[G_0] \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2$. In particular, $f : [G_0]^2 \to 2$ is stable. Since any $\Sigma^0_n(G_0 \oplus Y)$ formula is $\Sigma^0_{n-1}(P_0)$, and $\mathcal{M}[P_0] \models T_{n-1}$, $\mathcal{M}[G_0] \models T_n$. In particular, since $\mathsf{RCA}_0 + T_n \vdash \mathsf{WF}(\omega^\omega$, $\mathcal{M}[G_0] \models \mathsf{WF}(\omega^\omega)$.

One can see $f : [G_0]^2 \to 2$ as an instance of SEM. By Proposition 0.2 applied to $\mathcal{M}[G_0]$, P and $\alpha = 1$, there exists some M-infinite f-transitive set $G_1 \subseteq G_0$ such that $P \geqslant_T (G_1 \oplus G_0 \oplus Y)'$. Here again, since any $\Sigma^0_n(G_1 \oplus G_0 \oplus Y)$ formula is $\Sigma^0_{n-1}(P)$ and $\mathcal{M}[P] \models T_{n-1}$, $\mathcal{M}[G_1 \oplus G_0] \models T_n$.

One can see $f : [G_1]^2 \to 2$ as an instance of SADS. By Proposition 0.3 applied to $\mathcal{M}[G_0 \oplus G_1]$, P and $\alpha = 0$, there exists some M-infinite f-homogeneous set $G \subseteq G_1$ such that $P \geqslant_T (G \oplus G_1 \oplus G_0 \oplus Y)'$ and $\mathcal{M}[G \oplus G_1 \oplus G_0] \models \mathsf{RCA}_0 + \mathsf{B}\Sigma^0_2$. Still by the same argument, $\mathcal{M}[G \oplus G_1 \oplus G_0] \models T_n$. □

References

[1] Marta Fiori-Carones, Leszek Aleksander Kołodziejczyk, Tin Lok Wong, and Keita Yokoyama. An isomorphism theorem for models of Weak König's Lemma without primitive recursion. *Journal of the European Mathematical Society*, 2021. To appear.